

16. A body moves with simple harmonic motion of amplitude A and frequency $\frac{b}{2\pi}$.

What is the magnitude of the acceleration when the body is at maximum displacement?

- A** zero **B** $4\pi^2 Ab^2$ **C** Ab^2 **D** $\frac{4\pi^2 A}{b^2}$

Answer: C

$a = -\omega^2 x$ for SHM, so magnitude of acceleration $= \omega^2 x$

At maximum displacement A , acceleration $= (2\pi f)^2 A = \left(2\pi \times \frac{b}{2\pi}\right)^2 A = Ab^2$